Aspen Johnson

+1 (561) 299-8424 | aspenjohnson2024@gmail.com | linkedin.com/in/aspenjjohnson/ | github.com/DormantBillionaire |
www.aspenhq.dev

Education

Valdosta State University
Incoming B.S. in Data Science: Computational Science & Engineering Track; Minors: Physics and Mathematics

Palm Beach State College
A.A. in STEM: Mechanical & Aerospace Engineering Track — *Completed May 2026, GPA: 3.20*

Experience

Physics Undergraduate Researcher — Mar 2026 – Present
Valdosta State University — *On-Site/Remote*

- Collaborate with Professor S. Mubin on computational modeling research connecting financial-market illiquidity, time-series behavior, and larger particle simulation concepts.
- Use Python-based tools including NumPy, Matplotlib, Google Colab, NotebookLM, and Perplexity to support data analysis, visualization, literature review, and technical documentation.
- Review model outputs, visualizations, and research notes for accuracy, clarity, reproducibility, and consistency before faculty discussion.

Published Econometrics Researcher — Jan 2025 – Dec 2025
Mercer University — *On-Site/Remote*

- Co-authored *An empirical application of improved gradient scaling for score-driven volatility filters*, published in *Applied Economics Letters*, Taylor & Francis, 2025.
- Contributed to data curation, formal analysis, methodology, software implementation, validation, visualization, and manuscript preparation for empirical testing of the Hessian-filter scaled Beta-t-EGARCH volatility model.
- Analyzed financial time-series data across Bitcoin, S&P 500, Gold, U.S. REITs, and CHF/JPY from 2010–2025 using R, GAUSS, Excel, and Bloomberg Terminal data.
- Selected as 1 of 15 research papers to present a lecture at the 2026 Atlanta Research Conference.

Financial Analyst Intern — May 2025 – Dec 2025
Zinzino USA — *Remote*

- Reduced chargeback processing time by 87% through workflow digitization, documentation streamlining, and process-improvement analysis.
- Created reference documentation to support onboarding, data quality, process consistency, and cross-functional accuracy.
- Reviewed finance-related documentation and case details for completeness, accuracy, and clarity before sharing guidance with internal stakeholders.

Projects

Econometric Modeling in Complex Physical Systems | *Python, R, Time-Series Analysis, Volatility Modeling*

- Investigating whether GARCH-family and long-memory time-series models can be adapted to complex physical systems exhibiting clustering, persistence, uncertainty, or volatility-like behavior.
- Sourcing and structuring geophysical datasets from USGS, NOAA, and NCAR, including seismic activity, geomagnetic storms, and solar wind turbulence, for empirical model validation.
- Exploring how financial-market modeling tools may support broader analysis of event clustering, infrastructure-adjacent risk, system behavior, and reliability-focused interpretation.
- Documenting model assumptions, limitations, data quality issues, and interpretation risks to support reproducible research workflows and avoid overclaiming predictive results.

Assistive Hearing Technology Concept | *Accessibility Research, Signal Processing, Embedded Systems*

- Investigating a low-cost assistive hearing technology concept combining accessibility research, audio/signal processing concepts, embedded systems, and human-centered design.
- Researching how microphones, sensors, microcontrollers, and software-based filtering could support improved sound awareness in athletic events and high-noise environments.
- Documenting user needs, technical requirements, component options, and design constraints for future prototype development and applied technology testing.

DIY Cyberdeck / Portable Computing Hardware System | *Fusion 360, KiCad, PCB Design, Embedded Systems*

- Designing a modular portable computing device inspired by the ClockworkPi uConsole, emphasizing enclosure constraints, component layout, usability, repairability, and field-ready portability.

- Using Fusion 360 to prototype enclosure concepts, internal spacing, mounting points, and component placement for a compact hardware system.
- Learning KiCad workflows for schematic capture, PCB layout, component selection, and hardware design documentation.
- Applying electronics, electromagnetism, embedded systems, and hardware/software integration concepts to strengthen applied engineering and technical documentation skills.

Leadership, Community Engagement & Extracurriculars

Division I Women's College Basketball Player | *Mercer University, 2022–2025*

Division II Women's College Basketball Player | *Valdosta State University, Present*

Weekly Physics Study Group Member | *Studying Helliwell's Modern Classical Mechanics*

Read2Succeed Tutor | *Supported elementary literacy development and student confidence at Matilda Hartley Elementary*

AWS Campus Prep Series Participant | *Completed career and cloud-focused webinar series*

Technical Skills

Programming & Analysis: Python, R, GAUSS, Java, C/C++, JavaScript, Octave, Excel
Data Science & Modeling: Data cleaning, exploratory data analysis, time-series analysis, statistical modeling, model validation, data visualization, forecasting concepts
AI/Data Review: AI response evaluation, prompt testing, data annotation, data labeling, data quality review, pattern recognition
Engineering & Prototyping: Fusion 360, KiCad, PCB design fundamentals, circuit design fundamentals, embedded systems concepts, technical diagrams
Tools: NumPy, Matplotlib, Seaborn, Google Colab, LaTeX, Git, CLI, VS Code, Bloomberg Terminal
Research & Documentation: Literature review, technical writing, reproducible notes, model comparison, workflow documentation